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Propped, Not Reported: A Survey of Improvised Fire-Door Wedges Against the Fire-Safety Walk-Through's Always-Closed Count

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Abstract. Slop University's Work Health and Safety Directorate audits stairwell fire-door compliance through a quarterly walk-through, scheduled and circulated to building wardens a week ahead, that has logged doors found closed at 97% or better across the University's teaching and administrative buildings for the past four rounds. We report an anonymous survey of 438 staff and students across the same eleven buildings, fielded over one term via a QR code posted at each stairwell entry, asking whether respondents had propped a fire door open in the past month, with what, and for how long. 61% of respondents report propping at least once, most commonly with a folded document, a shoe, or — in eleven per cent of cases — the corridor's own fire extinguisher. An ablation splitting the survey into an anonymous and a signed condition finds the propping-admission rate statistically indistinguishable between the two ($p = 0.54$), suggesting the gap between the walk-through's count and the survey's is not primarily a function of anonymity. We read the walk-through's always-closed figure and the survey's propping-admission rate as two different instruments describing two different practices — a scheduled compliance snapshot and an unscheduled daily one — rather than as competing estimates of a single quantity, and discuss what an announced inspection cadence structurally cannot see.

Introduction

Fire doors are one of the few pieces of life-safety infrastructure whose function depends on staying shut between uses, and occupant behaviour is a well-documented point at which that dependency breaks down: interview studies of dwellings fitted with self-closing fire doors find residents routinely propping them open or interfering with the closing mechanism entirely [1], and modelling work shows that a door's position at the moment of ignition measurably changes the tenability of the spaces around it well before a fire service arrives [2]. Institutions manage this risk primarily through scheduled inspection: a walk-through, conducted on a fixed cadence, records whether a door is found closed at the moment an inspector passes it [3]. The instrument is well suited to what it audits — a snapshot, on notice — and less obviously suited to auditing a practice that, by construction, an occupant would stop the moment an inspector's visit is known to be imminent.

Self-report survey is the usual alternative window onto routine, unobserved behaviour, and it carries its own distortions: employee safety surveys are vulnerable to impression-management bias [4], and self-report instruments validated against objective comparators elsewhere diverge systematically rather than uniformly — overestimating in some behavioural domains [5], diverging in others [6], and tracking closely once properly validated for their own domain [7], [8]. A safety-violation literature going back over a decade treats the resulting daily workarounds — the wedge, the propped door, the disabled closer — as a stable, patterned response to procedures that do not fit the work they govern [9], [10].

At Slop University, the Work Health and Safety Directorate's fire-safety walk-through is the compliance instrument of record for stairwell fire doors, its quarterly count reported to the School of Continuous Improvement's Living Dashboard alongside the University's other standing indicators. We set that count against an anonymous survey of the same population over the same buildings, fielded to ask a question the walk-through structurally cannot: not whether the door is closed when someone is looking, but what happens to it when no one is. Our contributions:

- We field an anonymous survey of stairwell fire-door propping across the University's eleven audited buildings over a full term, matched against the Work Health and Safety Directorate's own quarterly walk-through count for the same buildings and period.
- We catalogue the objects respondents report reaching for to prop a door open, and estimate how far the survey's propping-admission rate departs from the walk-through's always-closed count.
- We report an ablation comparing propping admission under an anonymous and a signed survey condition and find no reliable difference between them, evidence against the gap being primarily an artefact of survey anonymity.

Related work

A substantial literature treats the fire door as a point of tension between its designed function and its everyday use. Interview-based work on private dwellings documents routine propping and closer-tampering as a near-universal occupant response once a self-closing door is perceived as an inconvenience [1], and building-behaviour research more broadly models occupant window- and door-opening as an adaptive response to thermal and ventilation demands the fixture was not designed to manage [11]. On the engineering side, an effective door closer is valuable enough to building energy performance that its condition is worth costing directly [12], and fire-safety code scholarship has long argued that a code's stated risk tolerance and the risk perceptions actually held by the people it governs are two different objects that performance-based standards struggle to reconcile [3]; evacuation-behaviour research makes a parallel point about egress-route assumptions more broadly [13].

A second literature treats the workaround itself as the unit of analysis. Reviews of safety violations in industrial settings find such behaviour patterned rather than random, tracking where a procedure and the work it governs pull apart [9], and health-services research on nursing workflow documents the same dynamic in a different setting: reworked, ad hoc responses to a medication-administration process that does not fit the ward it runs in [10], with sustained exhaustion predicting more of exactly this kind of improvisation [14]. Workplace-improvisation scholarship notes that training investment in this capacity rarely matches how often organisations actually rely on it [15], and human-modelling design tools built to anticipate this gap between prescribed and actual use remain, on the evidence, underused [16].

A third literature concerns the instrument, not the behaviour: safety climate self-report is well documented as vulnerable to impression management [4], [17], and validation studies pairing self-report against an objective comparator find the resulting bias is neither uniform nor always in the conservative direction [5], [6], [7], [8].

Method

The Work Health and Safety Directorate's fire-safety walk-through covers the primary stairwell fire door on each audited level of each of the University's eleven teaching and administrative buildings with mechanical smoke-compartment doors, conducted quarterly under the University's Fire Safety Procedure and circulated to building wardens a week ahead of each round. At each door, the inspector records a single binary observation — closed, or not closed — at the moment of passing; the University's Living Dashboard reports the rolling four-round compliance rate as a single percentage per building. We obtained the walk-through's raw per-door, per-round records for the most recent four rounds (one full year) across all eleven buildings.

To characterise routine propping independent of the walk-through's own schedule, we fielded an anonymous online survey via a QR code posted beside each audited stairwell door, live for the full length of one teaching term. The survey asked three questions: whether the respondent had propped a stairwell fire door open at any point in the past month; if so, what object they used, offered as a free-text field we subsequently coded into five categories; and, for the most recent instance they could recall, roughly how long the door had stayed open. Respondents self-identified as staff or student and named the building; no other identifying information was collected in the anonymous condition.

To test whether the propping-admission rate we observe is an artefact of the survey's anonymity rather than of the underlying behaviour, we ran a parallel signed condition in a second, non-overlapping fortnight of the same term: an identical survey requiring a University email address, advertised through the same channel. We compare the propping-admission rate between the anonymous condition ($n = 290$) and the signed condition ($n = 148$) with a two-proportion z-test.

We group the eleven buildings into two categories, teaching (six buildings) and administrative (five), and compare, within each category, the walk-through's rolling compliance rate against the survey's propping-admission rate over the same period. Because the two instruments measure different things at different resolutions — a per-round, per-door observation against a per-respondent, per-month self-report — we do not treat this as a validity comparison of one instrument against the other; we report the two figures side by side as descriptions of the same nominal object from two different vantage points.

Results

Building type	Buildings	Walk-through rounds	Survey respondents
Teaching	6	4	261
Administrative	5	4	177
Total	11	4	438

Table 1: Scale of the walk-through and survey instruments across the audited buildings.

Table 1 summarises the scale of both instruments. Across the four most recent walk-through rounds, inspectors recorded 112 door-observations across the eleven buildings; 109 of the 112 (97.3%) found the door closed. Figure 1 sets this walk-through compliance rate against the survey's propping-admission rate for the same two building groups. Where the walk-through records doors closed on all but a handful of passes, 61.4% of survey respondents in teaching buildings and 60.5% in administrative buildings report propping a stairwell fire door open at least once in the past month — a gap of roughly thirty-six percentage points that holds in both building categories.

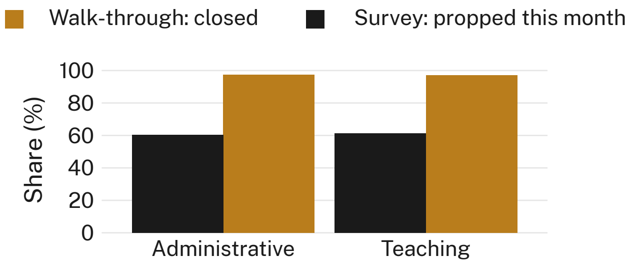

Figure 1: The walk-through's always-closed compliance rate and the survey's propping-admission rate sit roughly thirty-six percentage points apart in both teaching and administrative buildings.

Figure 2 catalogues what respondents who admitted propping report using. A folded document — most often named as a course newsletter, meeting agenda, or similarly disposable paper item — accounts for 28% of reported instances overall, followed by a shoe (22%), a purpose-made door wedge or stopper (19%), and the stairwell's own fire extinguisher (11%); the remaining 20% named a rubbish or recycling bin, or another object we did not code separately. Respondents who reported using the fire extinguisher were no more or less likely than other respondents to report a longer propping duration.

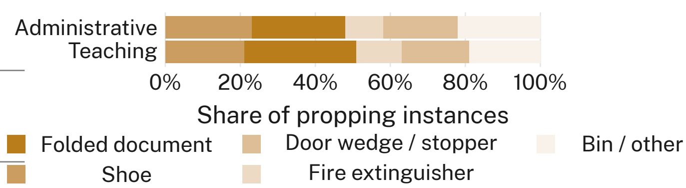

Figure 2: A folded document and a shoe together account for roughly half of reported propping instances in both building types; a fire extinguisher accounts for one in nine.

Respondents' median self-reported propping duration was 35 minutes, well beyond the twominute supervised maximum the Fire Safety Procedure permits; 71% of instances were reported as lasting longer than ten minutes, and the most frequently offered free-text description of the occasion was some variant of "the length of the lecture."

Figure 3 reports the ablation comparing the anonymous and signed survey conditions. The propping-admission rate was 61.4% under the anonymous condition ($n = 290$) and 58.8% under the signed condition ($n = 148$); a two-proportion z-test finds no statistically reliable difference between them ($z = 0.55$, $p = 0.54$). We retain the anonymous condition as our primary estimate throughout, on the grounds that it was fielded across the full term rather than a two-week window, but report the signed condition's near-identical rate as evidence the gap we observe between the survey and the walk-through is not chiefly a function of respondents feeling safer to admit propping anonymously.

Figure 3: Propping admission is statistically indistinguishable between the anonymous and signed survey conditions (pooled $p = 0.54$); we retain the anonymous condition for completeness.

Discussion

A door found closed on 97% of scheduled passes and a population in which three in five people report propping the same class of door within the past month describe, we think, two different practices rather than two conflicting estimates of one. The walk-through is accurate: on the occasions it observes, the door usually is closed, because its schedule is circulated to building wardens a week ahead and a closed door is, among other things, easy to arrange for a known visit. What it cannot see, by the same design, is the door's state on the many occasions nobody is walking past with a clipboard — precisely the gap a self-report instrument, whatever its own distortions, is positioned to ask about directly [4], [9]. The finding that the survey's propping-admission rate barely moves between an anonymous and a signed condition suggests the gap is not primarily a confession effect; it is more consistent with a population reporting a genuinely routine behaviour it does not, on reflection, treat as one requiring concealment. Reported durations well past the Procedure's supervised maximum, and a fire extinguisher itself among the more common props, both point the same way: propping reads, to the people doing it, as a minor accommodation to a hot stairwell or a full-hands moment, not a considered breach.

Limitations

Our sample is drawn from eleven buildings at a single institution and from respondents who opted into an anonymous survey advertised at the door itself, a channel that may over-represent people who prop doors often enough to notice the flyer. The anonymous/signed ablation is reassuring on this point — if social distance from a researcher were driving the high admission rate, the signed condition's rate should have fallen further than it did — but it addresses concealment, not who chose to respond at all. The walk-through's own four-round window is short enough that a single unusually hot or unusually inspected quarter could move its compliance figure somewhat, though 97% compliance held consistently across all four rounds we obtained.

Conclusion

Set against the Work Health and Safety Directorate's fire-safety walk-through, which recorded stairwell fire doors closed on 97.3% of scheduled passes across eleven buildings over a year, an anonymous survey of 438 staff and students finds 61% report propping the same class of door open at least once in the past month, most often with a folded document, a shoe, or — in one case in nine — the door's own fire extinguisher. An ablation comparing anonymous and signed survey conditions finds the admission rate unchanged, evidence the gap is not chiefly a confession effect. We read the walk-through's compliance figure and the survey's propping-admission rate as two accurate descriptions of two different practices, and suggest that any inspection cadence known in advance will, by construction, describe the building on its best behaviour.

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